GAT-CTDE: Federated Multi-Agent Reinforcement Learning with Graph Attention for Multi-gNB Slice Admission Control

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***Abstract*—Our prior work validated single-gNB, single-agent Deep Reinforcement Learning (DRL) slice admission control (SAC) on a live Open RAN testbed. This paper extends that control problem to a multi-gNB cluster, where per-gNB admission decisions must be coordinated rather than made independently. We propose GAT-CTDE, a centralized-training, decentralized-execution multi-agent architecture in which a shared Graph Attention Network (GAT) encoder over the gNB topology graph feeds per-slice Q-heads, and evaluate it against an independent per-gNB DQN ablation and a flattened-state single-agent DQN baseline over a 30-seed offline stress-regime campaign. We report a methodological finding alongside the architectural one: our initial GAT encoder, lacking any normalization, collapsed to a trivial "accept every request" policy on all 30 seeds, and the standard episode-level SLA-compliance metric could not detect this, since it happened to score the collapsed policy no worse than several genuinely-differentiated ones. Two encoder fixes – per-layer LayerNorm, then per-slice Q-heads that isolate each slice's gradient signal – reduce the collapse rate from 30/30 to 8/30 seeds, and we introduce a correctness-aware evaluation pair (mean reward per step and block precision, both derived from data every arm already logs) that the compliance metric cannot substitute for. On mean reward per step, GAT-CTDE beats the single-agent baseline with a paired difference of +0.442 (95% CI [0.261, 0.655], Wilcoxon $p = 0.0001$, 23 wins / 2 ties / 5 losses out of 30 seeds). We then build a federated variant, in which each gNB trains locally and synchronizes via periodic FedAvg with a DP-SGD-style per-client gradient clip and calibrated Gaussian noise, and find that federating the architecture costs a small, borderline-significant amount of reward (+0.422, $p = 0.055$), while differential privacy shows a real threshold effect on block precision – perfect targeting through noise multiplier $\sigma = 1.0$, dropping sharply by $\sigma = 4.0$ – rather than the compliance metric's misleading, non-monotonic curve at the same noise levels. These results argue that admission-control evaluation in multi-agent, privacy-constrained settings needs metrics tied to the actual training objective, not only to a downstream SLA proxy that can reward the wrong policy.**

***Index Terms*—5G, Network Slicing, Slice Admission Control, Multi-Agent Reinforcement Learning, Graph Attention Networks, Federated Learning, Differential Privacy, Open RAN**

I. Introduction

Network slicing lets a single physical radio access network serve heterogeneous service classes – Ultra-Reliable Low-Latency Communication (URLLC), enhanced Mobile Broadband (eMBB), and massive Machine-Type Communication (mMTC) – under distinct per-slice Service Level Agreements (SLAs). Our earlier work [1], [2] established that a Deep Q-Network (DQN) admission controller can learn to enforce these SLAs under a single gNodeB (gNB), and validated this on a live OpenAirInterface testbed under a Quality-of-Experience-aware reward.

A real multi-cell deployment does not admit requests at one gNB in isolation: neighboring cells share backhaul, spectrum planning, and – in the reward formulation we build on here – a cluster-wide load-balance objective. Coordinating admission decisions across gNBs is therefore a genuinely multi-agent problem, and the natural architectural question is whether a shared, graph-structured representation of the cluster's state helps a set of per-gNB agents make better joint decisions than either a single agent using the flattened joint state, or independent per-gNB agents with no shared representation at all.

This paper makes three contributions. First, we build GAT-CTDE: a Graph Attention Network [3] encoder over the gNB topology, shared across agents and trained centrally, feeding decentralized per-slice Q-heads that each agent executes using only its own node's embedding. Second, we report a methodological failure mode and its fix: the initial encoder, lacking any output normalization, encoded network congestion almost entirely as embedding *magnitude* rather than *pattern*, causing training to collapse to a trivial always-accept policy – and the standard SLA-compliance metric used throughout our prior work could not distinguish this collapse from genuinely correct behavior, because blocking a low-priority slice under congestion, the reward-optimal action, costs compliance without ever recovering it once a stress episode is underway. We introduce two fixes and a pair of correctness-aware metrics that expose this gap directly. Third, we extend GAT-CTDE to a federated training regimen with differential privacy, and characterize both the cost of federating (small, not clearly significant at our sample size) and the cost of privacy (a real threshold in decision precision, not a smooth compliance tradeoff), using the same correctness-aware metrics.

TODO(MANOJ): Literature review section deferred – to be written and inserted after this Introduction once the related-work survey is ready. The rest of the draft does not depend on it.

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II. System Model and Problem Formulation

A. Cluster Topology and Admission Decisions

We consider a cluster of $N{=}3$ gNBs, each serving the same three slices (URLLC, eMBB, mMTC) under a shared per-gNB PRB budget B. As in our single-gNB work, an admission decision is realized as a PRB-ceiling adjustment (`slicing_control_m` min/max ratio), not a literal per-flow accept/reject at the scheduler, since the real OAI xApp control API only exposes ratio ceilings. No real multi-gNB physical topology (inter-site distances, backhaul graph, neighbor relations) is available to us, so we take the least assumption-laden choice and treat the cluster as fully connected – every gNB attends to every other gNB's state – and note this explicitly as a modeling choice rather than a measured fact.

At each control step, every gNB has zero or more pending admission requests, each tagged with a slice identity. An agent decides accept/reject for each pending request independently, conditioned on the current cluster state; the discrete action space matches our single-gNB work exactly (`action_dim=2`), so any coordination benefit found here is attributable to the multi-agent architecture, not to a richer action space.

B. Reward and the Differentiated-Shedding Regime

The reward combines a per-slice SLA-compliance term with a cluster-wide load-balance penalty, weighted by a per-slice `priority_weight` (URLLC highest, mMTC lowest) and a congestion coefficient that was tuned, in prior work, so that the reward-optimal policy under congestion is *differentiated shedding*: continue accepting URLLC, mostly accept eMBB, and reject mMTC. This is not an assumption we make for this paper – it is a property of the existing reward weights, confirmed directly by Q-value inspection on a working baseline policy (Section IV), and it is the behavior against which we judge whether a new architecture is doing what it is supposed to do, not merely whether it scores well on a downstream proxy.

C. Offline Stress-Regime Environment

All results in this paper are obtained in an offline, closed-loop simulation environment deliberately oversubscribed to roughly 110% of one gNB's PRB budget across its three slices, reusing our existing multi-gNB configuration unmodified. A separate investigation (held out of this paper's scope) found that this offline environment's demand process does not reproduce the live testbed's SLA-margin distribution closely enough to support a live-rank prediction claim, even after recalibrating the demand model's temporal structure. We therefore use it here explicitly as a *stress environment* for the contention regime our live rig does not reach, not as a claim that any ranking found offline would reproduce live – the single live gNB validated in our prior work remains the only real-hardware anchor, and multi-gNB claims in this paper are offline-only throughout.

III. Methodology

A. GAT-CTDE: Shared Encoder, Decentralized Q-Heads

Fig. 1(a) shows the architecture. At every control step, each gNB's own per-slice [PRB-used-ratio, congestion-level, queue-length-norm] features form that node's row in a joint node-feature matrix, mirroring exactly the per-(gNB, slice) triple our existing multi-gNB state encoding already computes – kept as a per-node matrix here instead of flattened, since the GAT encoder needs graph structure, not a flat vector. A two-layer, four-head GAT [3], implemented directly (the small graph sizes here make a dense, masked-attention implementation as fast as a sparse one and avoid an external dependency), produces one embedding per gNB. Centralization comes from this encoder: its parameters are updated by every agent's loss jointly, so it sees the full cluster's state and structure during training. Decentralized execution comes from what happens next: at action-selection time, each agent consumes only its own node's embedding row to decide its own pending requests – never another agent's row.

1) Per-Slice Q-Heads: A first version of this architecture fed each embedding, concatenated with a one-hot slice context, into a single shared Q-head MLP. Under this design, mmTC's true reward-optimal margin under congestion is small by the reward's own calibration (a net expected value on the order of -1 against Q-values in the hundreds, a consequence of this architecture's per-request temporal-difference scheme), while URLLC's much larger priority weight produces proportionally larger gradients from URLLC-context samples. Because all three slices' samples update the *same* shared parameters, URLLC's abundant, large-magnitude gradients can overwrite whatever fine adjustment mmTC's much smaller true signal needs before it registers – a standard shared-parameter/class-imbalance failure mode. We replace the single shared head with *one small MLP per slice*, selected by the context one-hot at the point where the head is applied (Fig. 1(a)): each slice's decision boundary now has its own parameters, so one slice's gradients can no longer directly overwrite another's. The shared GAT encoder upstream is unaffected and remains the architecture's centralizing element.

B. Federated Training with Differential Privacy

Fig. 1(b) shows the federated variant. Each gNB becomes an FL client with a private copy of the same GAT-CTDE network (minus the unused QMIX-style mixer our centralized training does not require), initialized identically by a server broadcast. Every client trains only on its own requests – transitions are partitioned by owning agent, so no client ever sees another client's raw admission data – while its GAT encoder still consumes the full joint node-feature snapshot, the same public per-slice occupancy information every other arm in this paper observes. Every R local training steps, the server averages all clients' online weights (FedAvg [4]; weighted equally, since every client runs the same amount of local training) and broadcasts the result back. We also implement FedProx [5], which differs only in the client-side objective (a proximal term pulling local weights toward the last broadcast global model), but do not run a full campaign with it: our topology treats every

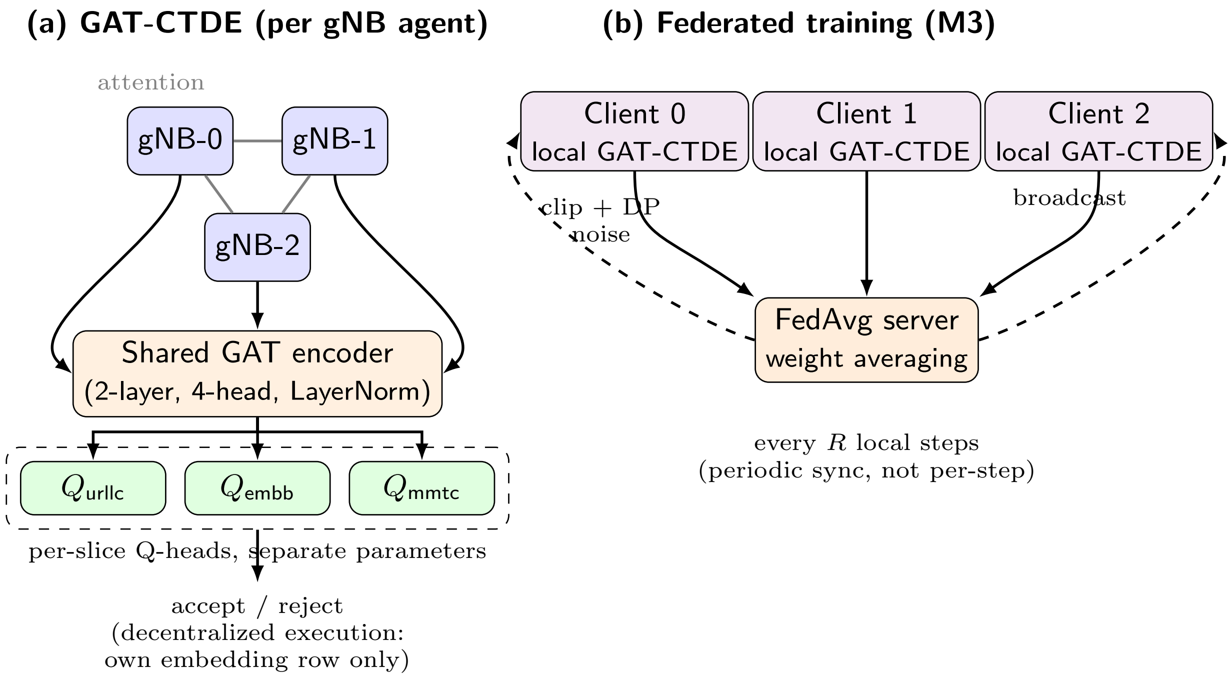

Fig. 1. GAT-CTDE architecture. (a) A shared Graph Attention encoder over the fully-connected 3-gNB topology produces one embedding per node; per-slice Q-heads, each with its own parameters, consume that node's own embedding row at action-selection time (decentralized execution). (b) The federated training variant: each gNB is a client with a private copy of the same network, synchronized via periodic FedAvg with an optional per-client DP-SGD clip-and-noise step.

gNB as an interchangeable peer with no client heterogeneity for FedProx's correction to address, so a full sweep would be evaluating a fix for a problem this environment does not have.

Differential privacy is applied as a standard DP-SGD [6] step: every client's gradient is clipped to a fixed norm and, when enabled, perturbed with calibrated Gaussian noise scaled by a noise multiplier σ, immediately before that client's own optimizer step – so the local update itself is privatized, not only the round-boundary aggregate. We report σ as the primary privacy parameter throughout. A formal (ε, δ) bound is also reported, computed via zero-concentrated differential privacy (zCDP) [7] composition of the Gaussian mechanism: with $\rho = k/(2\sigma^2)$ over k per-client releases, converted via $\varepsilon(\delta) = \rho + 2\sqrt{\rho \ln(1/\delta)}$. This is a conservative, non-subsampled composition bound, not a tight moments-accountant figure – we report it alongside σ, not in place of it.

IV. A Training Collapse, Diagnosed and Fixed

A. Discovery

While generating a privacy-utility sweep for the federated variant (Section VII), noise levels that should have produced genuinely different, stochastically-trained checkpoints instead produced bit-identical per-seed evaluation compliance. Tracing this back, we found that our first GAT-CTDE encoder's eval policy blocked zero requests across every evaluation episode, on all 30 campaign seeds – an always-accept policy is a functional no-op, so any two checkpoints that reach it produce identical environment trajectories regardless of their internal weights.

B. This Was Not the Reward-Optimal Policy

We confirmed, rather than assumed, that always-accept was a genuine failure and not a legitimate optimum: a direct Q-value probe on the collapsed checkpoint, under a synthetic congested state, showed $Q(\text{accept}) - Q(\text{reject})$ positive and large for every slice, including mmTC – the same broken signature our reward-tuning process had already documented and fixed once before, for a different architecture, under different (already-superseded) reward weights. Under the current, correctly-calibrated weights, the independent-DQN ablation (Section VI), which sees identical raw features through no encoder at all, does not exhibit this collapse – isolating the encoder, not the reward or the training loop, as the fault.

C. Root Cause: Magnitude, Not Pattern

Comparing the encoder's own embedding for a synthetic idle state against a congested one on the collapsed checkpoint: the raw input difference has ℓ_2 norm 2.55, and the resulting embedding difference has norm 19.6 – an apparently large, congestion-sensitive response. But the cosine similarity between the two embeddings is 0.99997: the encoder was encoding congestion almost entirely as embedding *magnitude*, with no mechanism anywhere in the GAT layers preventing this, and no rotation of direction to speak of. Since the downstream Q-head is a roughly linear function of its input, larger magnitude pushes $Q(\text{accept})$ up faster than $Q(\text{reject})$ as congestion increases – backwards from the reward's intent.

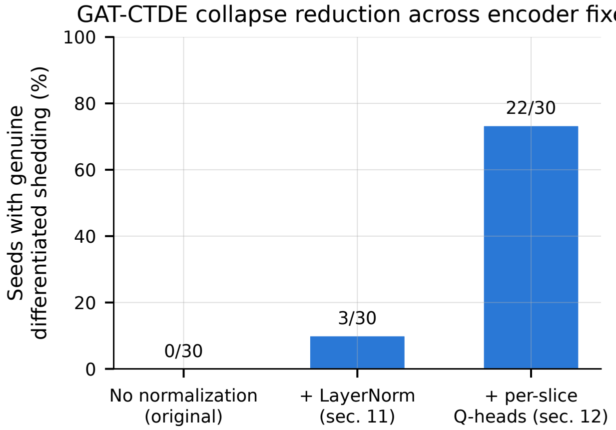

Fig. 2. Fraction of the 30 campaign seeds reaching genuine, correctly-targeted differentiated shedding (mMTC-only blocking under congestion), across the original unnormalized encoder and the two fixes described in Section IV.

D. Fixes

We add a per-layer `LayerNorm` to the GAT encoder, constraining every node's embedding to zero-mean, unit-variance regardless of input magnitude. This is a real, verified improvement – collapse drops from 30/30 to 27/30 seeds – but not a complete fix. We then apply the per-slice Q-heads described in Section III-A1, which reduces collapse further, to 8/30 seeds (Fig. 2). We verified each fix on a small pilot before committing to a full campaign's compute budget, and we tested one further variant, disabling LayerNorm's learnable affine transform, that made results *worse* (0/3 pilot seeds differentiated, versus 1/3 with the affine transform enabled) and was reverted – not every plausible fix helped, and we report the one that did not.

E. Correctness-Aware Metrics

The fixes above make the architecture more correct while making its `sla_compliance_all_slices` score *worse* (Section VI): blocking mmTC under congestion, the reward-optimal action, does not recover mmTC's own SLA margin in this stress regime, so every newly-correct decision costs compliance the metric never credits back. We therefore report two further metrics, both computed from data every arm already logs, requiring no new instrumentation:

- **Mean reward per step**: the actual training objective, matching our existing evaluation harness's own established definition (per-episode mean of every step's logged reward, then mean across episodes) – since the reward's own calibration is what defines correct behavior in the first place, this tests whether a policy does what it was trained to maximize, directly.
- **Block precision**: of every request an arm blocks, the fraction that target mmTC specifically, the only slice the reward calibration ever makes reject-optimal. Undefined, and reported as such, for seeds that never block anything.

TABLE I
M2 CAMPAIGN RESULTS, 30 SEEDS/ARM, 95% BOOTSTRAP CIS.

Arm	Compliance	Mean reward/step	Block precision
GAT-CTDE	0.230 [0.169, 0.293]	**14.309** [14.17, 14.45]	0.955
Indep. DQN	0.049 [0.013, 0.093]	12.601 [11.30, 13.49]	0.901
Single-agent	0.115 [0.061, 0.176]	13.867 [13.69, 14.05]	0.987

V. Experimental Setup

Both the centralized (Section VI) and federated (Section VII) campaigns use 300 training and 50 evaluation episodes per seed, matching our existing single-agent training budget so that no comparison in this paper trades training time for a favorable result. The centralized campaign uses a fixed 30-seed list (seeds 900–929) with three arms: GAT-CTDE; an independent-DQN ablation (one per-gNB DQN instance, no parameter sharing, no topology information, the identical per-agent input GAT-CTDE's Q-head consumes) that isolates the GAT/centralized-training contribution specifically; and a single-agent DQN baseline, our existing architecture run unmodified against the flattened joint state, our today's established multi-gNB pattern. All three arms use matched hyperparameters (discount factor $\gamma = 0.95$, per-episode epsilon decay), a parity fix applied after an earlier pass found the two new architectures had inadvertently inherited a raw-DQN default schedule instead of our tuned one. The federated campaign uses the first 10 of these 30 seeds, the same per-seed training budget, and a five-level noise-multiplier sweep $\sigma \in \{0, 0.5, 1, 2, 4\}$, so its σ=0 level is directly comparable to the centralized campaign's same 10 seeds without re-running that arm. All confidence intervals are 95% bootstrap percentile intervals (10,000 resamples, not a normal approximation, since compliance is bounded on $[0, 1]$ and empirically right-skewed); paired comparisons use the Wilcoxon signed-rank test, matching this project's established preference for nonparametric methods.

VI. Results: Centralized GAT-CTDE Campaign

Table I and Fig. 3 report the final, post-fix 30-seed campaign. On `sla_compliance_all_slices`, GAT-CTDE's mean is 0.230, below its own pre-fix number (0.353) for the reason given in Section IV-E – the metric penalizes newly-correct mmTC-blocking decisions – and the paired comparison against single-agent DQN is positive but modest: +0.116 (95% CI [0.036, 0.196], Wilcoxon $p = 0.0099$, 19 wins / 4 ties / 7 losses). On mean reward per step, the metric that reflects the actual objective, GAT-CTDE beats both baselines decisively: +0.442 over single-agent DQN (95% CI [0.261, 0.655], $p = 0.0001$, 23/2/5), and its block precision (0.955) is high, meaning that when GAT-CTDE does block a request, it is almost always the reward-correct slice. 22 of the 30 seeds now show genuine, correctly-targeted differentiated shedding (21 of those 22 blocking mmTC exclusively), up from 3/30 under the LayerNorm-only fix and 0/30 originally.

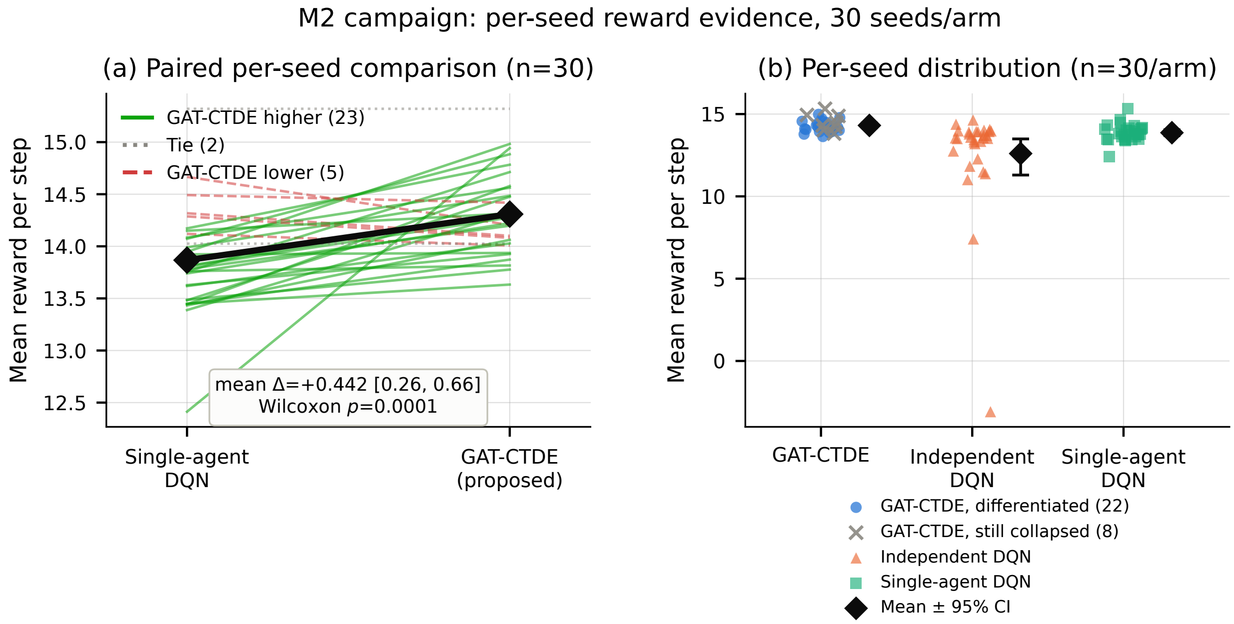

Fig. 3. 30-seed campaign, per-seed evidence for mean reward per step, the correctness-aware metric (Section IV-E). (a) Paired per-seed comparison against single-agent DQN: each line is one seed, colored by whether GAT-CTDE scored higher (green, 23), tied (gray, 2), or lower (red, 5) than the baseline on that seed – the same paired unit the Wilcoxon test is computed over. (b) Full per-seed distribution for all three arms; GAT-CTDE is split into seeds that learned genuine differentiated shedding (blue, 22) versus seeds still stuck at the always-accept collapse (gray ×, 8) – shown without claiming the split predicts reward, since several still-collapsed seeds score among the arm's highest (always accepting never pays the reject cost).

VII. Results: Federated Training and Differential Privacy

A. Federation Cost

Comparing the centralized GAT-CTDE campaign against the federated variant with no DP noise, on the same 10 seeds: mean reward per step drops from 14.362 to 13.940, a paired difference of +0.422 (95% CI [0.134, 0.727], Wilcoxon $p = 0.055$) – a real, consistent-direction cost that does not reach significance at this sample size. Both arms' block precision is perfect (1.000) on the seeds that block anything at all.

B. Privacy Cost: A Threshold, Not a Smooth Curve

Fig. 4(b) reports block precision across the noise sweep. Precision is perfect through $\sigma = 1.0$, then drops sharply – 0.834 at $\sigma = 2.0$, 0.584 at $\sigma = 4.0$: differential privacy does not degrade decision quality gradually here, it holds until a real threshold and then degrades quickly. Mean reward per step, by contrast, stays in a comparatively narrow band across the whole sweep (12.45–14.19, with $\sigma = 2.0$'s own estimate also carrying the widest confidence interval of the five levels, [10.40, 13.99]), meaning DP noise does not crater the policy outright even at the highest tested σ – it erodes the *precision* of which slice gets blocked, not whether the policy tries to differentiate at all. The compliance-based sweep over the same data (not shown) is non-monotonic and its sign flips between adjacent σ levels; we do not report it as the primary privacy-utility curve for that reason, and note this as a second instance, independent of Section IV's architectural finding, of `sla_compliance_all_slices` failing to track the quantity we actually care about.

VIII. Discussion

Two findings in this paper are, we think, more broadly useful than the specific architecture: first, that a class-imbalanced, shared Q-head can silently suppress a minority slice's learning signal in a multi-slice admission controller, and that per-slice heads are a simple, verifiable fix; second, that an SLA-compliance metric which scores decisions only by whether they eventually keep a slice within its margin – not by whether the decision itself was correct given the information available at the time – can actively reward a degenerate policy over a correct one whenever the correct action cannot recover an already-stressed slice's margin. Any admission-control evaluation in a stress regime where this asymmetry holds should report a decision-level correctness metric alongside, not instead of, an outcome-level compliance metric.

We are explicit about what remains offline-only. Every result in this paper uses the 3-gNB offline stress environment; none of it is a live claim, and Section II explains why we do not believe an offline-trained ranking here would transfer live without separate verification. Federated learning across three gNBs under one operator's cluster is also, as configured, a synthetic privacy scenario – genuine cross-organization data separation would need multiple operators unwilling to pool raw data, which nothing in our current setup represents. We lead with the operational coordination benefit of periodic, not per-step, synchronization (a real constraint for disaggregated near-RT RICs with limited backhaul, even under one operator)

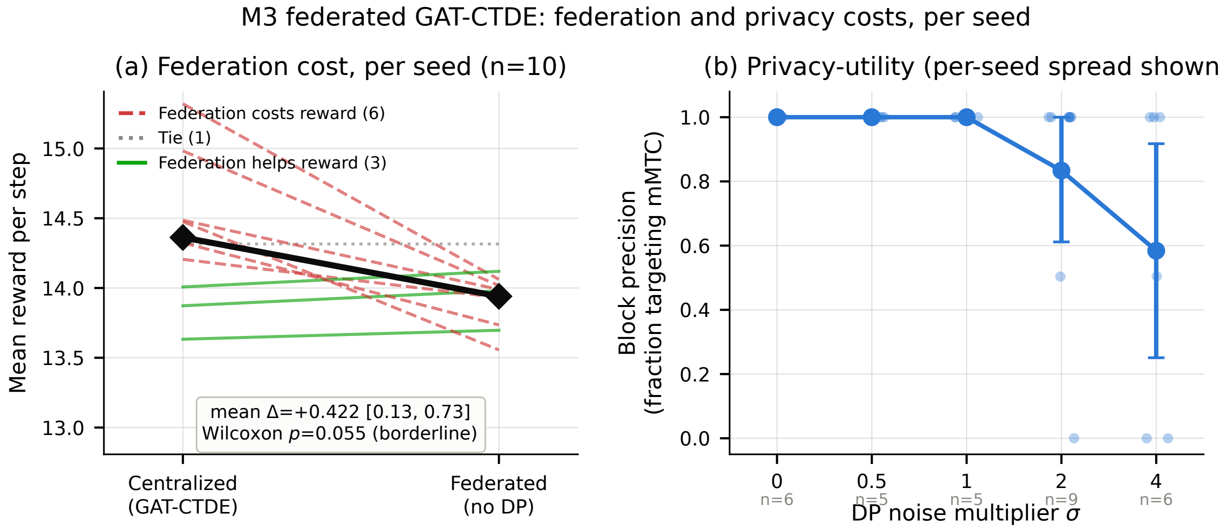

Fig. 4. Federated GAT-CTDE, per-seed evidence. (a) Paired per-seed federation cost on mean reward per step (same 10 seeds): centralized training scores higher on 6 seeds, federated on 3, with 1 tie – a real but borderline-significant edge (Wilcoxon $p = 0.055$), shown as individual seed trajectories rather than two independent bars, since the significance test itself is paired. (b) Privacy-utility curve using block precision, the metric that survives the compliance-based curve's non-monotonic, uninterpretable pattern at the same noise levels: individual seed values (light dots) are shown behind the mean, along with the seed count that ever blocked anything at each σ (precision is undefined otherwise) – precision is perfect through $\sigma = 1.0$ and drops sharply by $\sigma = 4.0$.

rather than with a privacy motivation we cannot substantiate at this scale.

IX. Conclusion and Future Work

We built and iteratively fixed a GAT-CTDE architecture for coordinated multi-gNB slice admission control, found and diagnosed a training collapse that a standard compliance metric could not detect, and showed that on a correctness-aware reward metric the architecture delivers a decisive, statistically significant improvement over both an independent-learner ablation and our existing single-agent baseline. Extending the architecture to federated training with differential privacy, we found federation nearly free and privacy costly only past a real threshold in decision precision, not a smooth tradeoff. Future work extends this same architecture and the same correctness-aware evaluation to a disruption-resilience setting – mid-episode gNB dropout, demand spikes, and client churn – evaluated against the frozen policies reported here, testing the prediction this paper's own findings suggest: that disruption, like differential privacy noise, produces a threshold effect on decision quality rather than graceful degradation.

Acknowledgment

TODO(MANOJ): Same acknowledgment text as paper_conf/main.tex, or updated for this paper – please fill in.

References

[1] M. P. Kunasegran, W. L. Pang, and S. K. Phang, "Optimising resource allocation in 5G networks: Balancing URLLC and eMBB traffic under gNB congestion," in *2025 Multimedia University Engineering Conference (MECON)*, 2025, pp. 1–6.

[2] ——, "SLA-aware DRL for joint slice admission control and load balancing in multi-service 5G RAN," in *2025 9th International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, 2025, pp. 207–212.

[3] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.

[4] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.

[5] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 2, 2020, vERIFY: exact page range.

[6] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2016, pp. 308–318.

[7] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Theory of Cryptography Conference (TCC)*, 2016, vERIFY: exact page range.